Example 2.32

Find the cube roots of unity.

Solution

We have to find $1^{\frac{1}{3}}$. Let $z = 1^{\frac{1}{3}}$ then $z^3 = 1$.

In polar form, the equation $z^3 = 1$ can be written as

$$z^3 = \cos(0 + 2k\pi) + i\sin(0 + 2k\pi) = e^{i2k\pi},\ k = 0,\ 1,\ 2, \cdots.$$

Therefore, $z = \cos\left(\frac{2k\pi}{3}\right) + i\sin\left(\frac{2k\pi}{3}\right) = e^{i\frac{2k\pi}{3}},\ k = 0,\ 1,\ 2$.

Taking $k = 0, 1, 2$, we get,

$k = 0,\quad z = \cos 0 + i\sin 0 = 1.$

$k = 1,\quad z = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = \cos\left(\pi - \frac{\pi}{3}\right) + i\sin\left(\pi - \frac{\pi}{3}\right)$

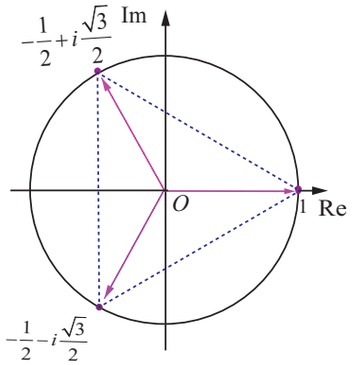
$$= -\cos\frac{\pi}{3} + i\sin\frac{\pi}{3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}.$$

$k = 2,\quad z = \cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3} = \cos\left(\pi + \frac{\pi}{3}\right) + i\sin\left(\pi + \frac{\pi}{3}\right)$

$$= -\cos\frac{\pi}{3} - i\sin\frac{\pi}{3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

Therefore, the cube roots of unity are

$$1, \frac{-1 + i\sqrt{3}}{2}, \frac{-1 - i\sqrt{3}}{2} \Rightarrow 1, \omega, \text{ and } \omega^2, \text{ where } \omega = e^{i\frac{2\pi}{3}} = \frac{-1 + i\sqrt{3}}{2}.$$



Cube roots of unity

Fig. 2.46

Example 2.33

Find the fourth roots of unity.

Solution

We have to find $1^{\frac{1}{4}}$. Let $z = 1^{\frac{1}{4}}$. Then $z^4 = 1$.

In polar form, the equation $z^4 = 1$ can be written as

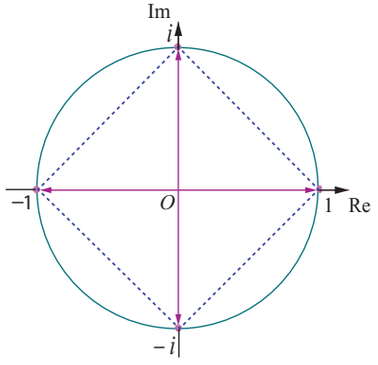
$$z^4 = \cos(0 + 2k\pi) + i\sin(0 + 2k\pi) = e^{i2k\pi}, k = 0,\ 1,\ 2, \ldots.$$

Therefore, $z = \cos\left(\frac{2k\pi}{4}\right) + i\sin\left(\frac{2k\pi}{4}\right) = e^{i\frac{2k\pi}{4}}$, $k = 0, 1, 2, 3$.

Taking $k = 0, 1, 2, 3$, we get

$k = 0,\quad z = \cos 0 + i\sin 0 = 1.$

$k = 1,\quad z = \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) = i.$



Fourth roots of unity

Fig. 2.47

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